THE SOLAR SYSTEM

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Contents

1 Our Planetary Neighborhood

1.1 The Sun and Its Planets

The Sun and its gravity-bound bodies form the Solar System[1]. See NASA Solar System Exploration. In increasing distance, the planets are

Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune

The planets form two groups:

- I **Inner Planets**
  - † Mercury, Venus, Earth, and Mars belong to this group.
  - † They are comparatively small and have solid surfaces.
  - † They are also called the terrestrial planets
- I **Outer Planets**
  - † Jupiter and Saturn are gas giants.
  - † Uranus and Neptune are commonly called ice giants.
  - † All four outer planets possess ring systems.

The inner planets are small, dense, and relatively close to the Sun, whereas the outer planets are much larger and travel along considerably wider orbits.

1.2 Portraits of the Solar System

1 shows the sun and eight planets. [1]

Compare Mars in 1e with Jupiter in 1f

2 Kepler's Laws of Planetary Motion

Kepler's three laws describe planetary motion[2].

2.1 First Law: Elliptical Orbits

A planetary orbit is elliptical, with the Sun at one focus:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \qquad a >= b > 0 \tag{1}$$

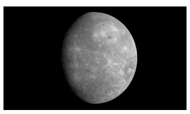
Here a and b are the semi-axes.

[1] Sizes and distances are schematic.

Star

(a) Country scale

Inner Planets



(b) Mercury

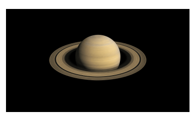
(c) Venus

(d) Earth

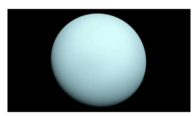
(e) Mars

Outer Planets

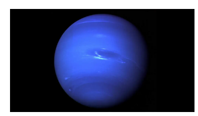
(f) Jupiter



(g) Saturn



(h) Uranus



(i) Neptune

Figure 1: The Sun and the eight planets of the Solar System, arranged into the inner and outer planetary groups. The displayed sizes and distances are not to scale.

For focal distance c,

$$\begin{aligned} c^2 &= a^2 + b^2 \\ e &= \frac{c}{a} \end{aligned} \tag{2}$$

Here e is the eccentricity.

In polar form,

$$r(\theta) = \frac{a(1-e^2)}{1+e\cos\theta} \tag{3}$$

Equation (3) gives the orbital distance.

2.2 Second Law: Equal Areas in Equal Times

Equal times sweep equal areas:

$$\Delta t_1 = \Delta t_2 \implies \Delta A_1 = \Delta A_2 \tag{4}$$

Consequently, a planet moves

- ⋄ faster when it is closer to the Sun, and
- ⋄ slower when it is farther from the Sun.

2.3 Third Law: Orbital Period and Distance

For planets orbiting one star,

$$T^2 \propto a^3. \tag{5}$$

Newtonian form:

$$T^2 = \frac{4\pi^3}{GM_\odot} a^3,$$

Here $M_\odot$ is the solar mass.

2.4 An Earth–Mars Comparison

Using Earth as reference,

$$a_E = 1AU \qquad\qquad T_E = 1\ year$$

where $1AU \approx 1.496 \times 10^{11}$. [2] For mars, $a_M = 1.524AU$, so

$$\begin{aligned} \frac{T_M^2}{T_E^2} &= \frac{a_M^3}{a_E^3} \\ T_M &= T_E \left(\frac{a_M}{a_E}\right)^{3/2} \\ &\approx (1.524)^{3/2} \approx 1.88 \text{ years.} \end{aligned}$$

Thus $T_M \approx 1.88$ years

3 A Relativistic Extension

General relativity extends Kepler's model.[3] Einstein's equations are

$$G_{\mu v} + Ag_{\mu v} = \frac{8\pi G}{c^4} T_{\mu v} \tag{6}$$

[2] AU denotes the mean Earth–Sun distance.
[3] Planetary corrections are small but measurable.

For spherical mass M,

$$
\begin{aligned}
ds^2 = -\left(1 - \frac{r_s}{r}\right) c^2 dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 \\
+r^2(d\theta^2 + \sin^2\theta d\phi^2), \qquad r_s = \frac{2GM}{c^2}
\end{aligned}
\tag{7}
$$

Free fall satisfies

$$
\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\alpha\beta}\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau} = 0 \tag{8}
$$

Per orbit

$$
\Delta\bar{\omega} = \frac{6\pi GM}{a(1-e^2)c^2} \tag{9}
$$

Proposition 1 (Weak-field circular orbit)**.** *For a circular orbit of radius r,*

$$
\left(\frac{v}{c}\right)^2 = \frac{GM}{rc^2} = \frac{r_s}{2r}. \tag{10}
$$

Proof. The circular-orbit balance $v^2/r = GM/r^2$ gives $v^2 = GM/r$. Divide by c^2 and use $r_s = 2GM/c^2$. □

See Proposition 1.

4 The Outer Planets

The outer planets are Jupiter, Saturn, Uranus, and Neptune[3]. The first two are gas giants; the last two are ice giants.

Table 1 uses scientific and relativistic notation.[4]

Table 1: Orbital parameters and relativistic notation for the outer planets.

Planet	**Orbital parameters**		**Relativistic notation**	
	a (AU)	e	$\varepsilon = \frac{1}{ac^2}GM_\odot$	$\beta = \sqrt{\varepsilon}$
Jupiter	5.204	4.89×10^{-2}	1.90×10^{-9}	4.36×10^{-5}
Saturn	9.583	5.65×10^{-2}	1.03×10^{-9}	3.21×10^{-5}
Uranus	19.191	4.72×10^{-2}	5.14×10^{-10}	2.27×10^{-5}
Neptune	30.070	8.6×10^{-3}	3.28×10^{-10}	1.81×10^{-5}

Table 1 compares the four gas giants

[4]Values are rounded for typesetting practice

References

[1] National Aeronautics and Space Administration. About the planets, 2026. Solar System Exploration.

[2] National Aeronautics and Space Administration. Orbits and kepler's laws, 2024.

[3] National Aeronautics and Space Administration. Solar system temperatures, 2022.